

Computer Networks(CS3001) Sessional-II Exam

Date: April 8th 2025

Course Instructor(s)

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Total Time (Hrs): **1**

Total Marks: **15**

Total Questions: **3**

Roll No

Section

Student Signature

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Attempt all the questions.

CLO # 1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Q1: Estimated Time: 15 minutes

[10 x 0.5 = 5 Marks]

Give your answer as true or false to show your agreement or disapproval to the following along with one sentence justification. Note: No points will be awarded without justification.

- a) Multiplexing and de-multiplexing occur only at the transport layer.
- b) In TCP, flow control ensures that the receiver does not overwhelm the sender.
- c) The UDP header includes a checksum field for error detection and correction.
- d) The maximum TCP segment size can be of 64KB only.
- e) TCP uses a cumulative acknowledgment strategy.
- f) The DNS client retransmits the query only after a timeout if no response is received.
- g) TCP protocol ensures that packets are never lost in the network.
- h) The TCP timeout value must be fixed for reliable communication.
- i) The transport layer is responsible for routing packets between networks.
- j) The sliding window protocol improves efficiency by allowing multiple packets to be in transit before receiving an acknowledgment.

Q2: Estimated Time: 20 minutes

[2.5 + 2.5 = 5 Marks]

Illustrate using the timing sequence diagram, the following:

- a) TCP 3-way Handshake
- b) Closing a TCP Connection

CLO # 2: Demonstrate the basics of network concepts using state-of-the-art network tools.

Q3: Estimated Time: 15 minutes

[2.5 + 2.5 = 5 Marks]

A sender transmits packets using a window size of 4 and a timeout interval of 10 ms. The sequence numbers available are 0, 1, 2, 3, 4, 5, and 6. Each packet takes 5 ms to be transmitted and acknowledged. The sender starts transmission at T = 0 ms. The timeout occurs at T = 10 ms.

Initially, the sender transmits packets 0, 1, 2, and 3. However, packet 1 is lost during transmission.

Draw time-sequence diagrams, for the given scenario, to illustrate the successful reception of the first four packets at the receiver, considering:

- a) Go-Back-N (GBN) protocol
- b) Selective Repeat (SR) protocol